

Problem 6 — Full Walkthrough

General rank-one matrix factorization

Setup: Let $A = x_0 y_0^T \neq 0$ be a rank-one matrix, $x_0, y_0 \in \mathbb{R}^n$ (note $A \neq 0$ forces *both* $x_0 \neq 0$ and $y_0 \neq 0$). Consider

$$f(x, y) := \frac{1}{2} \|A - xy^T\|_F^2, \quad x, y \in \mathbb{R}^n.$$

A fact we'll use throughout: as in Problem 5, $f(x, y) \geq 0$ always, with $f(x, y) = 0$ exactly when $xy^T = A$.

Step 1 — Expand $f(x, y)$ into a usable formula

Unlike Problem 5, xy^T is not symmetric in general, so use the fully general Frobenius identity $\|M\|_F^2 = \text{tr}(M^T M)$:

$$\|A - xy^T\|_F^2 = \text{tr}((A - xy^T)^T (A - xy^T)) = \text{tr}(A^T A) - \text{tr}(A^T xy^T) - \text{tr}((xy^T)^T A) + \text{tr}((xy^T)^T (xy^T)).$$

Handle each term. First, using the cyclic property of trace:

$$\text{tr}(A^T xy^T) = \text{tr}(y^T A^T x) = y^T A^T x = x^T A y$$

(the last step because a 1×1 matrix equals its own transpose: $y^T A^T x = (y^T A^T x)^T = x^T A y$). Similarly:

$$\text{tr}((xy^T)^T A) = \text{tr}(yx^T A) = \text{tr}(x^T A y) = x^T A y.$$

So together, $-\text{tr}(A^T xy^T) - \text{tr}((xy^T)^T A) = -2x^T A y$. For the last term:

$$(xy^T)^T (xy^T) = yx^T xy^T = (x^T x) yy^T = \|x\|^2 yy^T \implies \text{tr}((xy^T)^T (xy^T)) = \|x\|^2 \text{tr}(yy^T) = \|x\|^2 \|y\|^2.$$

Putting it together (and using $\text{tr}(A^T A) = \|A\|_F^2$):

$$f(x, y) = \frac{1}{2} \|A\|_F^2 - x^T A y + \frac{1}{2} \|x\|^2 \|y\|^2.$$

Specialize to $A = x_0 y_0^T$: $\|A\|_F^2 = \text{tr}(A^T A) = \text{tr}(y_0 x_0^T x_0 y_0^T) = \|x_0\|^2 \text{tr}(y_0 y_0^T) = \|x_0\|^2 \|y_0\|^2$. And $x^T A y = x^T x_0 y_0^T y = (x_0^T x)(y_0^T y)$. So:

$$f(x, y) = \frac{1}{2} \|x_0\|^2 \|y_0\|^2 - (x_0^T x)(y_0^T y) + \frac{1}{2} \|x\|^2 \|y\|^2.$$

Part (a) — Is the problem convex?

Test along $x = s u_0$, $y = s v_0$ (moving both variables together), where $u_0 := x_0/\|x_0\|$, $v_0 := y_0/\|y_0\|$ are unit vectors, and let $C := \|x_0\|\|y_0\| > 0$. Then $x_0^T x = s\|x_0\|$, $y_0^T y = s\|y_0\|$, $\|x\|^2 = \|y\|^2 = s^2$:

$$f(su_0, sv_0) = \frac{1}{2}C^2 - s^2C + \frac{1}{2}s^4.$$

This is exactly $\frac{1}{2}(C - s^2)^2$ (check: $\frac{1}{2}(C - s^2)^2 = \frac{1}{2}C^2 - Cs^2 + \frac{1}{2}s^4$ — matches). Differentiate twice:

$$\frac{d}{ds} \left[\frac{1}{2}(C - s^2)^2 \right] = -2s(C - s^2), \quad \frac{d^2}{ds^2} \left[\frac{1}{2}(C - s^2)^2 \right] = -2C + 6s^2.$$

At $s = 0$: the second derivative is $-2C < 0$. **So f is locally concave at $(0,0)$ along this line — f is not convex, and the problem is not a convex optimization problem.**

Part (b) — Find all stationary points

Step 1 — Compute the gradients

Work with $f(x, y) = \frac{1}{2}\|A\|_F^2 - x^T A y + \frac{1}{2}\|x\|^2\|y\|^2$ (general form). Note $x^T A y$ is *bilinear* (linear in x for fixed y , and linear in y for fixed x) — not quadratic like Problem 5's $x^T A x$, so no factor of 2 appears here.

Gradient of $x^T A y$ with respect to x : write $x^T A y = \sum_i \sum_j x_i A_{ij} y_j$. Treating y as fixed and differentiating with respect to x_k :

$$\frac{\partial}{\partial x_k} \left(\sum_i \sum_j x_i A_{ij} y_j \right) = \sum_j A_{kj} y_j = (A y)_k \implies \nabla_x (x^T A y) = A y.$$

Gradient with respect to y : differentiating the same sum with respect to y_k (now treating x fixed):

$$\frac{\partial}{\partial y_k} \left(\sum_i \sum_j x_i A_{ij} y_j \right) = \sum_i x_i A_{ik} = (A^T x)_k \implies \nabla_y (x^T A y) = A^T x.$$

Gradients of $\frac{1}{2}\|x\|^2\|y\|^2$: since $\|y\|^2$ is a constant with respect to x (and vice versa):

$$\nabla_x \left(\frac{1}{2}\|x\|^2\|y\|^2 \right) = \|y\|^2 x, \quad \nabla_y \left(\frac{1}{2}\|x\|^2\|y\|^2 \right) = \|x\|^2 y.$$

Combining:

$$\nabla_x f = -A y + \|y\|^2 x, \quad \nabla_y f = -A^T x + \|x\|^2 y.$$

Step 2 — Set both gradients to zero

$$A y = \|y\|^2 x \quad (\text{I}), \quad A^T x = \|x\|^2 y \quad (\text{II}).$$

With $A = x_0 y_0^T$ (so $A^T = y_0 x_0^T$): $A y = x_0 (y_0^T y)$ and $A^T x = y_0 (x_0^T x)$. So the system becomes:

$$(y_0^T y) x_0 = \|y\|^2 x \quad (\text{I}), \quad (x_0^T x) y_0 = \|x\|^2 y \quad (\text{II}).$$

Step 3 — Case $x = 0$

Equation (II) becomes $0 = \|0\|^2 y = 0$, automatically true regardless of y . Equation (I) becomes $(y_0^T y) x_0 = \|y\|^2 \cdot 0 = 0$. Since $x_0 \neq 0$, this forces $y_0^T y = 0$, i.e. $y \perp y_0$ (this includes $y = 0$ as a special case). **Stationary points:** $\{(0, y) : y \perp y_0\}$.

Step 4 — Case $y = 0$

By the symmetric argument (swap the roles of the two equations): (I) is automatic, and (II) forces $x_0^T x = 0$. **Stationary points:** $\{(x, 0) : x \perp x_0\}$.

Step 5 — Case $x \neq 0$ and $y \neq 0$

Since $y \neq 0$, $\|y\|^2 \neq 0$, so equation (I) can be solved explicitly for x :

$$x = \frac{y_0^T y}{\|y\|^2} x_0.$$

So x is automatically a scalar multiple of x_0 ; call the scalar $\lambda := (y_0^T y)/\|y\|^2$, so $x = \lambda x_0$. Since we're in the case $x \neq 0$ and $x_0 \neq 0$, we need $\lambda \neq 0$.

Since $x \neq 0$, $\|x\|^2 \neq 0$, so equation (II) can likewise be solved explicitly for y :

$$y = \frac{x_0^T x}{\|x\|^2} y_0.$$

Substitute $x = \lambda x_0$: $x_0^T x = \lambda \|x_0\|^2$ and $\|x\|^2 = \lambda^2 \|x_0\|^2$, so

$$y = \frac{\lambda \|x_0\|^2}{\lambda^2 \|x_0\|^2} y_0 = \frac{1}{\lambda} y_0.$$

So for every $\lambda \neq 0$, the pair $(x, y) = (\lambda x_0, y_0/\lambda)$ satisfies both (I) and (II) — direct substitution confirms this closes the loop consistently. **Stationary points:** $\{(\lambda x_0, y_0/\lambda) : \lambda \neq 0\}$.

Notice: $xy^T = (\lambda x_0)(y_0/\lambda)^T = \lambda \cdot \frac{1}{\lambda} \cdot x_0 y_0^T = x_0 y_0^T = A$ *exactly*, for every one of these points. So this entire family achieves $f = 0$.

Conclusion of Part (b)

Stationary points: $\{(0, y) : y \perp y_0\} \cup \{(x, 0) : x \perp x_0\} \cup \{(\lambda x_0, y_0/\lambda) : \lambda \neq 0\}$.

Part (c) — Show every local optimum is a global optimum

Step 1 — The third family are global minima

Every point in $\{(\lambda x_0, y_0/\lambda) : \lambda \neq 0\}$ satisfies $xy^T = A$ exactly (Step 5 above), so $f = 0$ there — the smallest value f can take anywhere (since $f \geq 0$ always). **These are global minima**, hence automatically local minima too.

Step 2 — The degenerate families are not local minima

Take a point $(0, \bar{y})$ with $\bar{y} \perp y_0$ (this covers the first family; the second is symmetric). Its value:

$$f(0, \bar{y}) = \frac{1}{2} \|x_0\|^2 \|\bar{y}\|^2 - 0 + 0 = \frac{1}{2} \|x_0\|^2 \|\bar{y}\|^2 > 0$$

(strictly positive, since $x_0, y_0 \neq 0$) — not the global minimum value 0, so not a global minimum. We show it isn't even a *local* minimum, by exhibiting a joint perturbation of both x and y that strictly decreases f .

Perturb $x = \varepsilon x_0$, $y = \bar{y} + \delta y_0$, for small ε, δ . Compute each piece of f :

$$x_0^T x = \varepsilon \|x_0\|^2, \quad y_0^T y = y_0^T \bar{y} + \delta \|y_0\|^2 = \delta \|y_0\|^2$$

(using $y_0^T \bar{y} = 0$). So the middle term of f :

$$(x_0^T x)(y_0^T y) = \varepsilon \delta \|x_0\|^2 \|y_0\|^2.$$

For the last term of f : $\|x\|^2 = \varepsilon^2 \|x_0\|^2$, and

$$\|y\|^2 = \|\bar{y} + \delta y_0\|^2 = \|\bar{y}\|^2 + 2\delta(y_0^T \bar{y}) + \delta^2 \|y_0\|^2 = \|\bar{y}\|^2 + \delta^2 \|y_0\|^2$$

(again using $y_0^T \bar{y} = 0$). So:

$$f(\varepsilon x_0, \bar{y} + \delta y_0) = \frac{1}{2} \|x_0\|^2 \|y_0\|^2 - \varepsilon \delta \|x_0\|^2 \|y_0\|^2 + \frac{1}{2} \varepsilon^2 \|x_0\|^2 (\|\bar{y}\|^2 + \delta^2 \|y_0\|^2).$$

Step 3 — Choose δ proportional to ε to force a decrease

Set $\delta = k\varepsilon$ for a constant k to be chosen. Substituting:

$$f(\varepsilon x_0, \bar{y} + k\varepsilon y_0) = \frac{1}{2} \|x_0\|^2 \|y_0\|^2 + \underbrace{\varepsilon^2 \left[-k \|x_0\|^2 \|y_0\|^2 + \frac{1}{2} \|x_0\|^2 \|\bar{y}\|^2 \right]}_{\text{coefficient of } \varepsilon^2} + \underbrace{\frac{1}{2} \varepsilon^4 k^2 \|x_0\|^2 \|y_0\|^2}_{O(\varepsilon^4), \text{ negligible for small } \varepsilon}.$$

Choose k large enough that the bracketed coefficient is negative:

$$-k \|x_0\|^2 \|y_0\|^2 + \frac{1}{2} \|x_0\|^2 \|\bar{y}\|^2 < 0 \iff k > \frac{\|\bar{y}\|^2}{2 \|y_0\|^2}$$

— always achievable, just pick such a k (e.g. $k = \|\bar{y}\|^2 / \|y_0\|^2 + 1$). With this k fixed, for sufficiently small $\varepsilon > 0$ the $\varepsilon^2 \times$ (negative) term dominates the $O(\varepsilon^4)$ term, so:

$$f(\varepsilon x_0, \bar{y} + k\varepsilon y_0) < \frac{1}{2} \|x_0\|^2 \|y_0\|^2 = f(0, \bar{y}).$$

This exhibits points arbitrarily close to $(0, \bar{y})$ where f is strictly smaller — so $(0, \bar{y})$ is not a local minimum, for any $\bar{y} \perp y_0$ (including $\bar{y} = 0$). The symmetric argument (swap the roles of x, y and x_0, y_0) shows every point $(\bar{x}, 0)$ with $\bar{x} \perp x_0$ is likewise not a local minimum.

Conclusion of Part (c)

Every stationary point found in Part (b) falls into one of two categories: the family $\{(\lambda x_0, y_0/\lambda) : \lambda \neq 0\}$, which are all global minima (Step 1); or the degenerate families $\{(0, y) : y \perp y_0\} \cup \{(x, 0) : x \perp x_0\}$, which are not local minima at all (Step 2–3), so they don't count as “local optima” that could fail to be global. Since every point that actually qualifies as a local minimum is already a global minimum:

Every local optimum of this problem is a global optimum. ■